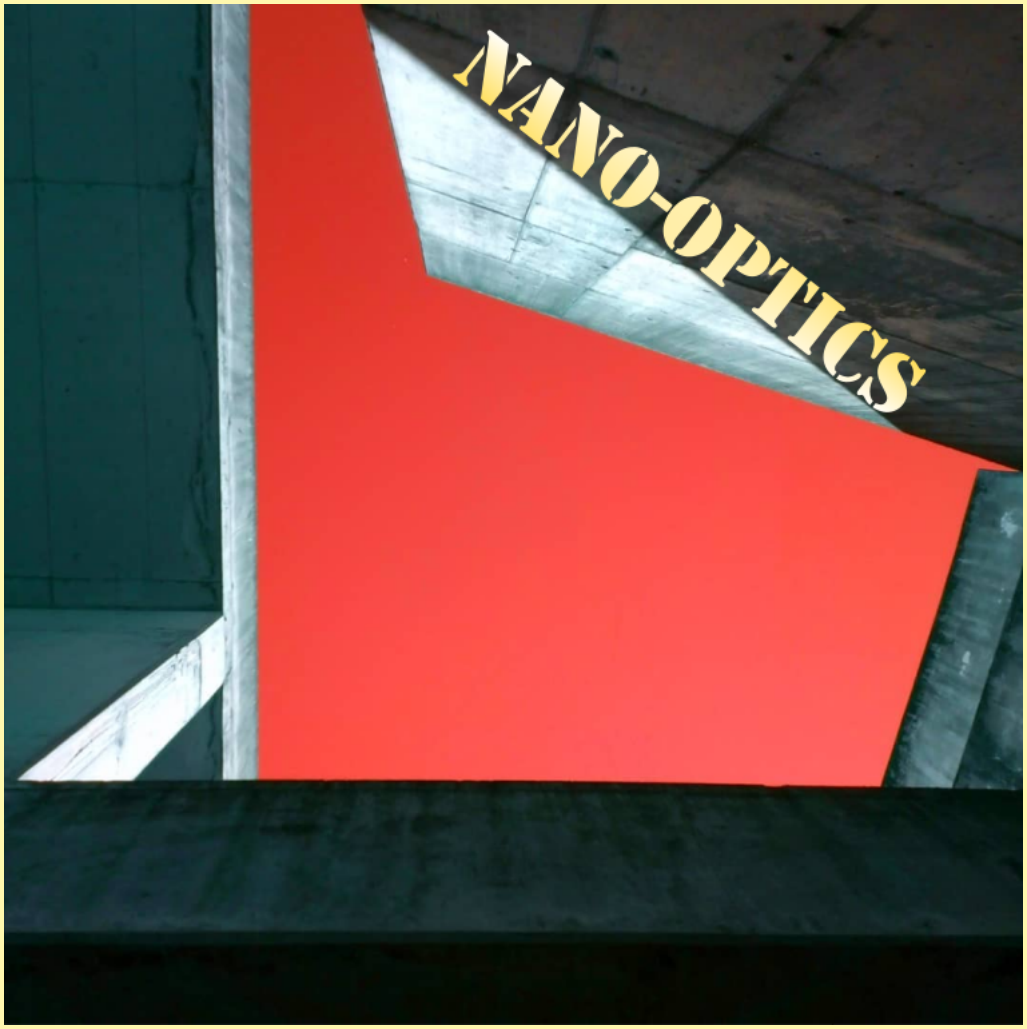
Plane Wave Expansion Methods for the Determination of Photonic Crystal Band Structures



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I. INTRODUCTION

In this work a discussion is given of the plane wave method applied for the determination of the electromagnetic modes and dispersion relation of a 2D photonic crystal. In its application the electromagnetic fields and function forms of the periodic dielectric are expressed in terms of Fourier expansion in plane waves. These expansion are then used to reduce the differential form of the wave equations, generated from the Maxwell equation, to a matrix eigenvalues problems. The matrix eigenvalue problems determine the Fourier coefficients of the plane wave expansion of the electromagnetic modes, relating them to the eigenfrequencies making up the dispersion relation of the system.

II. ELECTROMAGNETIC WAVES

A. Maxwell Equation

In SI units, the macroscopic Maxwell equations are of the form

$$\text{Gauss' law} \qquad \nabla \cdot \mathbf{D} = \rho_{\text{ext}} \tag{II.1}$$

$$\text{No Magnetic Monopoles} \qquad \nabla \cdot \mathbf{B} = 0 \tag{II.2}$$

$$\text{Faraday's law} \qquad \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0 \tag{II.3}$$

$$\text{Ampere-Maxwell's law} \qquad \nabla \times \mathbf{H} - \frac{\partial \mathbf{D}}{\partial t} = \mathbf{j}_{\text{ext}} \tag{II.4}$$

where $\mathbf{E} \equiv \mathbf{E}(\mathbf{r}, t)$ and $\mathbf{B} \equiv \mathbf{B}(\mathbf{r}, t)$ are the electric and magnetic fields, and $\rho_{\text{ext}} \equiv \rho_{\text{ext}}(\mathbf{r}, t)$ and $\mathbf{j}_{\text{ext}} \equiv \mathbf{j}_{\text{ext}}(\mathbf{r}, t)$ are the external/free charge and current densities, which act as sources of the fields. In a material medium, the response to an applied electric and magnetic fields is described by the $\mathbf{D} \equiv \mathbf{D}(\mathbf{r}, t)$ and $\mathbf{H} \equiv \mathbf{H}(\mathbf{r}, t)$ fields. These are defined as

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} \tag{II.5}$$

$$\mathbf{B} = \mu_0 \mathbf{H} + \mathbf{M} \tag{II.6}$$

where $\mathbf{P} \equiv \mathbf{P}(\mathbf{r}, t)$ and $\mathbf{M} \equiv \mathbf{M}(\mathbf{r}, t)$ are the polarization and magnetization of the medium.

If we now focus on the particular case of **monochromatic fields** fo frequency ω,

$$\mathbf{E}(\mathbf{r}, t) = \Re\left\{\mathbf{E}(\mathbf{r}) e^{-i\omega t}\right\} \quad \text{and} \quad \mathbf{B}(\mathbf{r}, t) = \Re\left\{\mathbf{B}(\mathbf{r}) e^{-i\omega t}\right\} \tag{II.7}$$

the Maxwell equations for the amplitudes and sources become

$$\nabla \cdot \mathbf{D}(\mathbf{r}) = \rho_{\text{ext}} \tag{II.8}$$

$$\nabla \cdot \mathbf{B}(\mathbf{r}) = 0 \tag{II.9}$$

$$\nabla \times \mathbf{E}(\mathbf{r}) - i\omega \mathbf{B}(\mathbf{r}) = 0 \tag{II.10}$$

$$\nabla \times \mathbf{H}(\mathbf{r}) + i\omega \mathbf{D}(\mathbf{r}) = \mathbf{j}_{\text{ext}}(\mathbf{r}) \tag{II.11}$$

If we assume a **local and isotropic response within the linear response regime**, this quantities are related to the applied fields according to the constitutive relations

$$\mathbf{P}(\mathbf{r}) = \varepsilon_0 \chi_e(\mathbf{r}, \omega) \mathbf{E}(\mathbf{r}) \tag{II.12}$$

$$\mathbf{M}(\mathbf{r}) = \mu_0 \chi_m(\mathbf{r}, \omega) \mathbf{H}(\mathbf{r}) \tag{II.13}$$

We note that the electric susceptibility χ_e and magnetic susceptibility χ_m are generally frequency-dependent, as a consequence of temporal dispersion. In *in-homogeneous media*, they also depend on position. The macroscopic fields are then given by

$$\mathbf{D}(\mathbf{r}) = \varepsilon_0 \varepsilon(\mathbf{r}, \omega) \mathbf{E}(\mathbf{r}) \tag{II.14}$$

$$\mathbf{B}(\mathbf{r}) = \mu_0 \mu(\mathbf{r}, \omega) \mathbf{H}(\mathbf{r}) \tag{II.15}$$

where the dielectric permittivity $\varepsilon(\mathbf{r}, \omega)$ and the magnetic permeability $\mu(\mathbf{r}, \omega)$ are defined as

$$\varepsilon(\mathbf{r}, \omega) = 1 + \chi_e(\mathbf{r}, \omega) \tag{II.16}$$

$$\mu(\mathbf{r}, \omega) = 1 + \chi_m(\mathbf{r}, \omega) \tag{II.17}$$

B. Helmholtz equation

Assuming a **linear, isotropic, non-magnetic media** (i.e $\mu(\mathbf{r},\omega) = 1$) and incident **monochromatic fields**, Maxwell equations read

$$\nabla \cdot \varepsilon_0 \left(\varepsilon(\mathbf{r},\omega)\mathbf{E}(\mathbf{r})\right) = \rho(\mathbf{r}), \tag{II.18}$$
$$\nabla \cdot \mathbf{B}(\mathbf{r}) = 0, \tag{II.19}$$
$$\nabla \times \mathbf{E}(\mathbf{r}) - i\omega\mathbf{B}(\mathbf{r}) = 0, \tag{II.20}$$
$$\nabla \times (i\omega)\mathbf{B}(\mathbf{r}) - k_0^2\varepsilon(\mathbf{r},\omega)\mathbf{E}(\mathbf{r}) = i\omega\mu_0\mathbf{j}(\mathbf{r}) \tag{II.21}$$

If we take the curl of Faraday's law,

$$\nabla \times \{\nabla \times \mathbf{E} - \nabla \times (i\omega)\mathbf{B} = 0\}, \tag{II.22}$$

and eliminate $\mathbf{B}$ using the Maxwell-Ampere's law, we obtain the *vector Helmholtz equation*

$$\nabla \times \nabla \times \mathbf{E}(\mathbf{r}) - k_0^2\varepsilon(\mathbf{r},\omega)\mathbf{E}(\mathbf{r}) = +i\omega\mu_0\mathbf{j}(\mathbf{r}) \tag{II.23}$$

In the **absence of external currents**, $\mathbf{j}(\mathbf{r}) = 0$, and assuming a **non-dispersive medium**, $\varepsilon(\mathbf{r},\omega) = \varepsilon(\mathbf{r})$, the vector Helmholtz equation reduces to

$$\boxed{\nabla \times \nabla \times \mathbf{E}(\mathbf{r}) = \frac{\omega^2}{c^2}\varepsilon(\mathbf{r})\mathbf{E}(\mathbf{r}) \qquad \text{(II.24)}}$$

where we used that, in free space, the wave vector relates with its frequency as $k_0^2 = \omega^2/c^2$.

III. 2D PHOTONIC CRYSTAL WITH 2D DISCRETE TRANSLATION SYMMETRY AND 1D CONTINUOUS TRANSLATION SYMMETRY

Consider a 2D photonic crystal with 2D discrete translation symmetry along the $\hat{\mathbf{x}}$- and $\hat{\mathbf{y}}$-direction but that is confined in the $\hat{\mathbf{z}}$-direction from $z = 0$ to $z = d$, as shown in Fig (1). Such a photonic crystal can be constructed as a crystal square lattice array of dielectric cylinders located between two parallel perfect conduction plates. The square lattice of the arrays is defined so that the location of the dielectric cylinder axes are at the lattice sites of the square lattice, and between the parallel plates and surrounding the dielectric cylinders is a vacuum background. The electromagnetic modes are confined in the region between the parallel plates where they exhibit a series of pass and stop band due to the periodicity of the array of the dielectric cylinders. Also, the dielectric cylinders do not overlap one another so that the radii of the cylinders, R, satisfy the condition $2R < a$.

1. *Helmholtz Equation*

Given all the consideration expressed in the section above, the Helmholtz equation, as derived, reads

$$\nabla \times \nabla \times \mathbf{E}(\mathbf{r}) = \frac{\omega^2}{c^2}\varepsilon(\mathbf{r})\mathbf{E}(\mathbf{r}) \tag{III.1}$$

In this case the dielectric function depends only on $\mathbf{r}_{||}$ and thus $\varepsilon(\mathbf{r}) = \varepsilon(\mathbf{r}_{||})$.

If we consider a **normal incidence on the crystal plane $x\mathcal{O}z$ or $y\mathcal{O}z$** such that $k_z = 0$, and, if we also consider the **electric field components parallel to the conducting plates to be zero**, then, in this limit, the boundary conditions at the parallel perfect conducting plates,

$$\mathbf{E}_{||}(\mathbf{r}_{||}, z = 0 \text{ or } z = d) = 0, \tag{III.2}$$

are trivially satisfied. For all intents and purposes, the resulting solution of the system, considering this mode, is identical with that of a photonic crystal composed of **infinite parallel dielectric cylinders**.

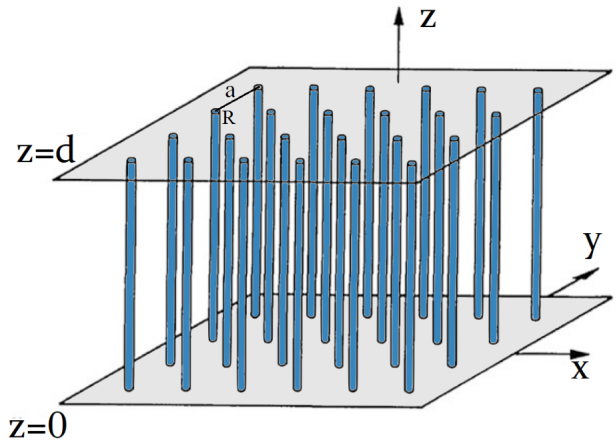

Figure 1. Schematic figure of the photonic crystal square lattice array of dielectric cylinders located between two parallel perfect conducting plates.

With this huge simplification the electric field reduced to

$$\mathbf{E}(\mathbf{r}) = \big(0, 0, E_z(\mathbf{r}_{||})\big) \tag{III.3}$$

where E_z does not on z anymore, only on $\mathbf{r}_{||}$. If we substitute (III.3) into (III.1), only a single component will survive. To see this, we invoke the relation

$$\nabla \times \nabla \times \mathbf{E}(\mathbf{r}) = \nabla\left(\nabla \cdot \mathbf{E}(\mathbf{r})\right) - \nabla^2 \mathbf{E}(\mathbf{r}) \tag{III.4}$$

The first term on the right is zero since

$$\nabla\left(\nabla \cdot \mathbf{E}(\mathbf{r})\right) = \nabla\left(\frac{\partial}{\partial x}0 + \frac{\partial}{\partial y}0 + \frac{\partial}{\partial z}E_z(\mathbf{r}_{||})\right) = 0 \tag{III.5}$$

while the second term on the right reads

$$\nabla^2 \mathbf{E}(\mathbf{r}) = \nabla^2_{\mathbf{r}_{||}} E_z(\mathbf{r}_{||}) + \frac{\partial^2}{\partial z^2} E_z(\mathbf{r}_{||}) = \nabla^2_{\mathbf{r}_{||}} E_z(\mathbf{r}_{||}) \tag{III.6}$$

Thus, our Helmholtz vector equation reduces to

$$\boxed{-\frac{1}{\varepsilon(\mathbf{r}_{||})}\nabla^2_{\mathbf{r}_{||}} E_z(\mathbf{r}_{||}) = \frac{\omega^2}{c^2} E_z(\mathbf{r}_{||}) \qquad \text{(III.7)}}$$

As we will explain briefly, given the symmetry considerations, one could immediately assume a general solution as

$$\boxed{\begin{aligned} E_z(\mathbf{r}_{||}) &= e^{i\mathbf{k}_{||}\cdot\mathbf{r}_{||}} u(\mathbf{r}_{||}) && \text{(III.8)} \\ \text{with} \quad u(\mathbf{r}_{||}) &= u(\mathbf{r}_{||} + \mathbf{T}_{||}) && \text{(III.9)} \end{aligned}}$$

Note for the general case:
Even more generally, without any of the considerations, the solution for any angle of incidence in a truncated 2D photonic crystal with 2D discrete translation symmetry, would still have the form of a plane wave times a function which has the periodicity of the given lattice, this is

$$E_i(\mathbf{r}_{||}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_i(\mathbf{r}_{||}, z) \tag{III.10}$$

$$u(\mathbf{r}_{||}, z) = u(\mathbf{r}_{||} + \mathbf{T}_{||}, z), \quad \text{for } i = x, y, z \tag{III.11}$$

In the z-direction there is no periodicity in the system but the fields satisfies the correct electromagnetic boundaries at the perfect conducting plates.

2. Symmetry considerations

In this considerations, the positions of the lattice sites of the square lattice in the xy-plane are given by 2D position vectors of the form

$$\mathbf{T}_{||} = an_x\hat{\mathbf{x}} + an_y\hat{\mathbf{y}}, \qquad n_x, n_y \in \mathbb{Z} \tag{III.12}$$

where a is the nearest neighbor separation between lattice points and n_1 and n_2 are integers. The corresponding reciprocal space vector $\mathbf{G}$, also defined in the xy-plane, is then given by

$$\mathbf{G}_{||} = \frac{2\pi}{a}n_x\hat{\mathbf{x}} + \frac{2\pi}{a}n_y\hat{\mathbf{y}}, \qquad n_x, n_y \in \mathbb{Z} \tag{III.13}$$

The reciprocal vector $\mathbf{G}$ has this form such that, for a generic periodic function,

$$f(\mathbf{r}_{||}) = f(\mathbf{r}_{||} + \mathbf{T}_{||}) \tag{III.14}$$

with $\mathbf{r}_{||}$ being the xy-in-plane position vector and its Fourier series expansion reading

$$f(\mathbf{r}_{||}) = \sum_{\mathbf{G}_{||}} f_{\mathbf{G}_{||}} e^{i\mathbf{G}_{||}\cdot\mathbf{r}_{||}} \tag{III.15}$$

we obtain

$$\begin{aligned} f(\mathbf{r}_{||}) = f(\mathbf{r}_{||} + \mathbf{T}) &\Leftrightarrow \sum_{\mathbf{G}_{||}} f_{\mathbf{G}_{||}} e^{i\mathbf{G}_{||}\cdot\mathbf{r}_{||}} = \sum_{\mathbf{G}_{||}} f_{\mathbf{G}_{||}} e^{i\mathbf{G}_{||}\cdot(\mathbf{r}_{||}+\mathbf{T}_{||})} \\ &\Rightarrow e^{i\mathbf{G}_{||}\cdot\mathbf{T}_{||}} = 1 \\ &\Rightarrow \mathbf{G}_{||}\cdot\mathbf{T}_{||} = 2\pi m, \quad \text{with } m \in \mathbb{Z} \end{aligned} \tag{III.16}$$

which, in fact, holds given the definitions (III.12) and (III.13).

Thus, the electric field function u, which necessarily has the property $u(\mathbf{r}_{||}) = u(\mathbf{r}_{||} + \mathbf{T}_{||})$ since it must obey the symmetry of the system, can then described by the Fourier expansion

$$u(\mathbf{r}_{||}) = \sum_{n_x,n_y} u_{n_x,n_y} e^{i\mathbf{G}_{n_x,n_y}\cdot\mathbf{r}_{||}} \tag{III.17}$$

Furthermore, the dielectric function $\varepsilon(\mathbf{r})$, which also has inherently the symmetry $\varepsilon(\mathbf{r}_{||}) = \varepsilon(\mathbf{r}_{||} + \mathbf{T}_{||})$, can the also be defined in reciprocal space as

$$\frac{1}{\varepsilon(\mathbf{r}_{||})} = \sum_{n_x^\varepsilon,n_y^\varepsilon} \kappa_{n_x^\varepsilon,n_y^\varepsilon} e^{i\mathbf{G}_{n_x^\varepsilon,n_y^\varepsilon}\cdot\mathbf{r}_{||}} \tag{III.18}$$

where the Fourier coefficients $\kappa_{n_x^\varepsilon,n_y^\varepsilon}$ can be found via an inverse Fourier transform, this is

$$\kappa_{n_x^\varepsilon,n_y^\varepsilon} = \frac{1}{A}\int_{\mathcal{A}} \frac{1}{\varepsilon(\mathbf{r}_{||})} e^{-i\mathbf{G}_{n_x^\varepsilon,n_y^\varepsilon}\cdot\mathbf{r}_{||}} d\mathbf{r}_{||} \tag{III.19}$$

where $\mathcal{A}$ goes over the entire photonic crystal. Soon we will derive the form of the coefficients $\kappa_{n_x^\varepsilon,n_y^\varepsilon}$ but, for now, let us continue with the symmetry considerations.

Given all the approximations we have done, specifically, normal incidence and $E_x(\mathbf{r}) = E_y(\mathbf{r}) = 0$ we are now dealing instead with infinite parallel dielectric cylinders and thus, there is no need to consider an extra dimension. For all effects our system is now 2D and can been seen as show in Fig (2).

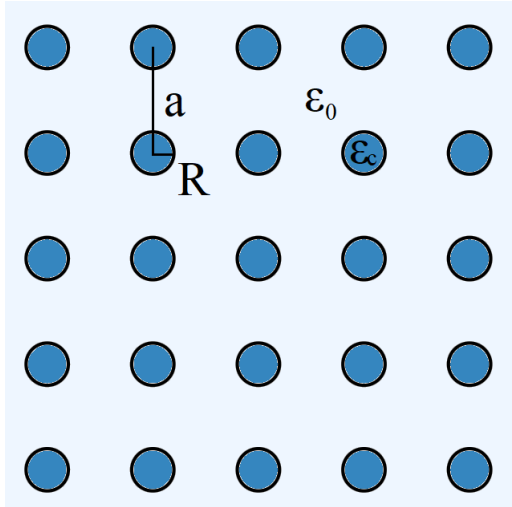

Figure 2. Schematic figure of the normal cross section of a photonic crystal square lattice array of infinite dielectric cylinders

3. *Finding the Solution*

Substituting (III.17), (III.18) into the differential equation (III.7) yields

$$-\left(\sum_{n_x^\varepsilon,n_y^\varepsilon}\kappa_{n_x^\varepsilon,n_y^\varepsilon}e^{i\mathbf{G}_{n_x^\varepsilon,n_y^\varepsilon}\cdot\mathbf{r}_{||}}\right)\nabla^2_{\mathbf{r}_{||}}\left(e^{i\mathbf{k}_{||}\cdot\mathbf{r}_{||}}\sum_{n_x,n_y}u_{n_x,n_y}e^{i\mathbf{G}_{n_x,n_y}\cdot\mathbf{r}_{||}}\right)=\frac{\omega^2}{c^2}\left(e^{i\mathbf{k}_{||}\cdot\mathbf{r}_{||}}\sum_{n_x,n_y}u_{n_x,n_y}e^{i\mathbf{G}_{n_x,n_y}\cdot\mathbf{r}_{||}}\right)$$

$$\Leftrightarrow+\sum_{n_x^\varepsilon,n_y^\varepsilon}\sum_{n_x,n_y}\kappa_{n_x^\varepsilon,n_y^\varepsilon}\left(\mathbf{G}_{n_x,n_y}+\mathbf{k}_{||}\right)^2e^{i(\mathbf{G}_{n_x^\varepsilon,n_y^\varepsilon}+\mathbf{k}_{||}+\mathbf{G}_{n_x,n_y})\cdot\mathbf{r}_{||}}u_{n_x,n_y}=\frac{\omega^2}{c^2}\sum_{n_x,n_y}e^{i(\mathbf{k}_{||}+\mathbf{G}_{n_x,n_y})\cdot\mathbf{r}_{||}}u_{n_x,n_y}\tag{III.20}$$

If we now introduce $n_x+n_x^\varepsilon=m_x\Leftrightarrow n_x^\varepsilon=m_x-n_x$ and $n_y+n_y^\varepsilon=m_y\Leftrightarrow n_y^\varepsilon=m_y-n_y$, given the definition of the reciprocal vectors in (III.13), we have that

$$\begin{aligned}G_{n_x^\varepsilon,n_y^\varepsilon}+G_{n_x,n_y}&=G_{m_x-n_x,m_y-n_y}+G_{n_x,n_y}\\&=\left(\frac{2\pi}{Na}(m_x-n_x)\hat{\mathbf{x}}+\frac{2\pi}{Na}(m_y-n_y)\hat{\mathbf{y}}\right)+\left(\frac{2\pi}{Na}n_x\hat{\mathbf{x}}+\frac{2\pi}{Na}n_y\hat{\mathbf{y}}\right)\\&=\frac{2\pi}{Na}\left((m_x-n_x)+n_x\right)\hat{\mathbf{x}}+\frac{2\pi}{Na}\left((m_y-n_y)+n_y\right)\hat{\mathbf{y}}\\&=\frac{2\pi}{Na}m_x\hat{\mathbf{x}}+\frac{2\pi}{Na}m_y\hat{\mathbf{y}}\\&=G_{m_x,m_y}\end{aligned}\tag{III.21}$$

and thus, we have instead

$$\Leftrightarrow\sum_{m_x,m_y}\sum_{n_x,n_y}\kappa_{m_x-n_x,m_y-n_y}\left(\mathbf{k}_{||}+\mathbf{G}_{n_x,n_y}\right)^2e^{i(\mathbf{k}_{||}+\mathbf{G}_{m_x,m_y})\cdot\mathbf{r}_{||}}u_{n_x,n_y}=\frac{\omega^2}{c^2}\sum_{n_x,n_y}e^{i(\mathbf{k}_{||}+\mathbf{G}_{n_x,n_y})\cdot\mathbf{r}_{||}}u_{n_x,n_y}\tag{III.22}$$

Since the summation indices are mute, we can say that $n_x\to m_x$ and $n_y\to m_y$ in the right-hand side such that the exponential terms cancel out and we are left with

$$\sum_{m_x,m_y}\left\{\sum_{n_x,n_y}\kappa_{m_x-n_x,m_y-n_y}\left(\mathbf{k}_{||}+\mathbf{G}_{n_x,n_y}\right)^2u_{n_x,n_y}=\frac{\omega^2}{c^2}u_{m_x,m_y}\right\}\tag{III.23}$$

The summation over m_x and m_y is now irrelevant since it has to be true for all individual terms and thus we can drop it. To make this matrix symmetric we introduce a new variable

$$u_{n_x,n_y}=\frac{v_{n_x,n_y}}{|\mathbf{k}_{||}+\mathbf{G}_{n_x,n_y}|}\tag{III.24}$$

such that

$$\sum_{n_x,n_y} \kappa_{m_x-n_x,m_y-n_y} \left(\mathbf{k}_{||}+\mathbf{G}_{n_x,n_y}\right)^2 \frac{v_{n_x,n_y}}{|\mathbf{k}_{||}+\mathbf{G}_{n_x,n_y}|} = \frac{\omega^2}{c^2}\frac{v_{m_x,m_y}}{|\mathbf{k}_{||}+\mathbf{G}_{m_x,m_y}|}$$

$$\Leftrightarrow \sum_{n_x,n_y} \kappa_{m_x-n_x,m_y-n_y}|\mathbf{k}_{||}+\mathbf{G}_{n_x,n_y}|v_{n_x,n_y}|\mathbf{k}_{||}+\mathbf{G}_{m_x,m_y}| = \frac{\omega^2}{c^2}v_{m_x,m_y} \tag{III.25}$$

And thus, the equation we ought to solve for the band dispersion is the eigenfrequencies equation

$$H^{m_x,m_y}_{n_x,n_y} v^{n_x,n_y} = \frac{\omega^2}{c^2} v_{m_x,m_y} \tag{III.26}$$

written in Einstein notation with the matrix elements being

$$H^{m_x,m_y}_{n_x,n_y} = |\mathbf{k}_{||}+\mathbf{G}_{n_x,n_y}|\kappa_{m_x-n_x,m_y-n_y}|\mathbf{k}_{||}+\mathbf{G}_{m_x,m_y}| \tag{III.27}$$

4. *Analytical Evaluation of $\kappa_{m_x-n_x,m_y-n_y}$ coefficients:*

Before we try to solve (III.26) we need the expression for the coefficients $\kappa_{m_x-n_x,m_y-n_y}$. This could be done numerically but the analytical solution is also simple enough to treat. Equation (III.19) tells us that:

$$\kappa_{n_x^\varepsilon,n_y^\varepsilon} = \frac{1}{A}\int_{\mathcal{A}} \frac{1}{\varepsilon(\mathbf{r}_{||})} e^{-i\mathbf{G}_{n_x^\varepsilon,n_y^\varepsilon}\cdot\mathbf{r}_{||}} d\mathbf{r}_{||} \tag{III.28}$$

Since the Dirac delta function has the Fourier transform

$$\delta(n_x^\varepsilon=0,n_y^\varepsilon=0) = \int_{\mathcal{A}} e^{-i\mathbf{G}_{n_x^\varepsilon,n_y^\varepsilon}\cdot\mathbf{r}_{||}} d\mathbf{r}_{||} \tag{III.29}$$

we can write (III.28) as

$$\kappa_{n_x^\varepsilon,n_y^\varepsilon} = \frac{1}{A}\int_{\mathcal{A}} \left[\frac{1}{\varepsilon(\mathbf{r}_{||})}-1\right] e^{-i\mathbf{G}_{n_x^\varepsilon,n_y^\varepsilon}\cdot\mathbf{r}_{||}} d\mathbf{r}_{||} + \delta(n_x^\varepsilon=0,n_y^\varepsilon=0) \tag{III.30}$$

The area of the photonic crystal A of the xy-plane can be broken into a set of identical smallest area units, a_c, containing a single lattice site of the photonic crystal lattice. (An example of such a division is to break the plane up into little squares centered about each cylinder so that the area A is obtained as sum of the little squares.) For this division of the plane it then follows that $A = Na_c$ with N being the number of lattice sites. With this notations (III.30) becomes

$$\kappa_{n_x^\varepsilon,n_y^\varepsilon} = \frac{1}{a_c}\int_{a_c} \left[\frac{1}{\varepsilon(\mathbf{r}_{||})}-1\right] e^{-i\mathbf{G}_{n_x^\varepsilon,n_y^\varepsilon}\cdot\mathbf{r}_{||}} d\mathbf{r}_{||} + \delta(n_x^\varepsilon=0,n_y^\varepsilon=0) \tag{III.31}$$

where the integral is over a single a_c centered at the origin of the xy-plane, and the periodicity of the integrand has been used.

The integral in (III.31) can be evaluated with the help of the Jacobi–Anger expansion (proven in appendix)

$$e^{-i\mathbf{G}_{n_x^\varepsilon,n_y^\varepsilon}\cdot\mathbf{r}_{||}} = \sum_{m=-\infty}^{+\infty} (-i)^m J_m(G_{n_x^\varepsilon,n_y^\varepsilon} r_{||}) e^{-im\theta} \tag{III.32}$$

where θ is the angle such that $\mathbf{G}_{n_x^\varepsilon,n_y^\varepsilon}\cdot\mathbf{r}_{||} = G_{n_x^\varepsilon,n_y^\varepsilon} r_{||}\cos\theta$. Substituting into (III.31) we obtain

$$\kappa_{n_x^\varepsilon,n_y^\varepsilon} = \frac{1}{a_c}\int_{a_c} \left[\frac{1}{\varepsilon(\mathbf{r}_{||})}-1\right] \left(\sum_{m=-\infty}^{+\infty} (-i)^m J_m(G_{n_x^\varepsilon,n_y^\varepsilon} r_{||}) e^{-im\theta}\right) d\mathbf{r}_{||} + \delta(n_x^\varepsilon=0,n_y^\varepsilon=0) \tag{III.33}$$

where the integrals can be now be easily evaluated as

$$
\begin{aligned}
&\frac{1}{a_c}\int r_{||}dr_{||}d\theta\left[\frac{1}{\varepsilon(\mathbf{r}_{||})}-1\right]\left(\sum_{m=-\infty}^{+\infty}(-i)^m J_m(G_{n_x^\varepsilon,n_y^\varepsilon}r_{||})e^{-im\theta}\right)\\
=&\frac{1}{a_c}\sum_{m=-\infty}^{+\infty}(-i)^m\left(\int_0^{2\pi}e^{-im\theta}d\theta\right)\left(\int_0^R r_{||}dr_{||}\left[\frac{1}{\varepsilon(r_{||})}-1\right]J_m(G_{n_x^\varepsilon,n_y^\varepsilon}r_{||})\right)\\
=&\frac{1}{a_c}\sum_{m=-\infty}^{+\infty}(-i)^m\left(2\pi\delta_{m,0}\right)\left(\int_0^R r_{||}dr_{||}\left[\frac{1}{\varepsilon(r_{||})}-1\right]J_m(G_{n_x^\varepsilon,n_y^\varepsilon}r_{||})\right)\\
=&\frac{1}{a_c}2\int_0^R r_{||}dr_{||}\left[\frac{1}{\varepsilon(r_{||})}-1\right]J_0(G_{n_x^\varepsilon,n_y^\varepsilon}r_{||})
\end{aligned}
\tag{III.34}
$$

Since $\varepsilon(r_{||})$ inside the dielectric cross section area a_c is constant ε_c, we have that

$$
\kappa_{n_x^\varepsilon,n_y^\varepsilon}=\left[\frac{1}{\varepsilon_c}-1\right]\frac{1}{a_c}2\pi\int_0^R r_{||}dr_{||}J_0(G_{n_x^\varepsilon,n_y^\varepsilon}r_{||})+\delta(n_x^\varepsilon=0,n_y^\varepsilon=0)
\tag{III.35}
$$

where, once again, ε_c is the dielectric constant of the cylinders of the photonic crystal. Now, applying the Bessel function identity

$$
\frac{d}{dx}\left(xJ_1(x)\right)=xJ_0(x)
\tag{III.36}
$$

by making the chance of variable $x=G_{n_x^\varepsilon,n_y^\varepsilon}r_{||}$ meaning that $r_{||}=x/G_{n_x^\varepsilon,n_y^\varepsilon}$, $dr_{||}=dx/G_{n_x^\varepsilon,n_y^\varepsilon}$, gives the result

$$
\begin{aligned}
\kappa_{n_x^\varepsilon,n_y^\varepsilon}&=\left[\frac{1}{\varepsilon_c}-1\right]\frac{1}{a_c}2\pi\frac{1}{(G_{n_x^\varepsilon,n_y^\varepsilon})^2}\int_0^{G_{n_x^\varepsilon,n_y^\varepsilon}R}dx\left(xJ_0(x)\right)+\delta(n_x^\varepsilon=0,n_y^\varepsilon=0)\\
&=\left[\frac{1}{\varepsilon_c}-1\right]\frac{1}{a_c}2\pi\frac{1}{(G_{n_x^\varepsilon,n_y^\varepsilon})^2}\left(G_{n_x^\varepsilon,n_y^\varepsilon}R\right)J_1(G_{n_x^\varepsilon,n_y^\varepsilon}R)+\delta(n_x^\varepsilon=0,n_y^\varepsilon=0)\\
&=\left[\frac{1}{\varepsilon_c}-1\right]\frac{\pi R^2}{a_c}\frac{2J_1(G_{n_x^\varepsilon,n_y^\varepsilon}R)}{G_{n_x^\varepsilon,n_y^\varepsilon}R}+\delta(n_x^\varepsilon=0,n_y^\varepsilon=0)
\end{aligned}
\tag{III.37}
$$

We now define the filling fraction of the cylinders in the plane of the photonic crystal plates, known as the *form factor,* as $f=\pi R^2/a_c$. The final result for the $\kappa_{n_x^\varepsilon,n_y^\varepsilon}$ terms yields

$$
\kappa_{n_x^\varepsilon,n_y^\varepsilon}=\frac{1}{\varepsilon_c}f+(1-f),n_x^\varepsilon\neq 0\text{ and }n_y^\varepsilon\neq 0
\tag{III.38}
$$

$$
\kappa_0=\left[\frac{1}{\varepsilon_c}-1\right]f\frac{2J_1(G_{n_x^\varepsilon,n_y^\varepsilon}R)}{G_{n_x^\varepsilon,n_y^\varepsilon}R},n_x^\varepsilon=0\text{ and }n_y^\varepsilon=0
\tag{III.39}
$$

To check if this analytical result is correct, one could evaluate numerically the dielectric functions as the Fourier series of this $\kappa_{n_x^\varepsilon,n_y^\varepsilon}$ coefficients using (III.18) to see if we obtain once again our dielectric function as an array of circle, as intended. In Fig (3) we show the results of exactly that calculation for a grid of $a=1$ of uniform dielectric cylinder of radii $R=0.2$ and $\varepsilon_c=3$ surrounded by vacuum $\varepsilon_0=1$.

5. *Numerical Evaluation*

Before we show the results of the numerical evaluation, we first show an example of how to implement said numerical implementation in *Julia* while also explaining the complications along the way.

Naturally we start be defining all the functions necessary to the problem as shown in Fig (4).

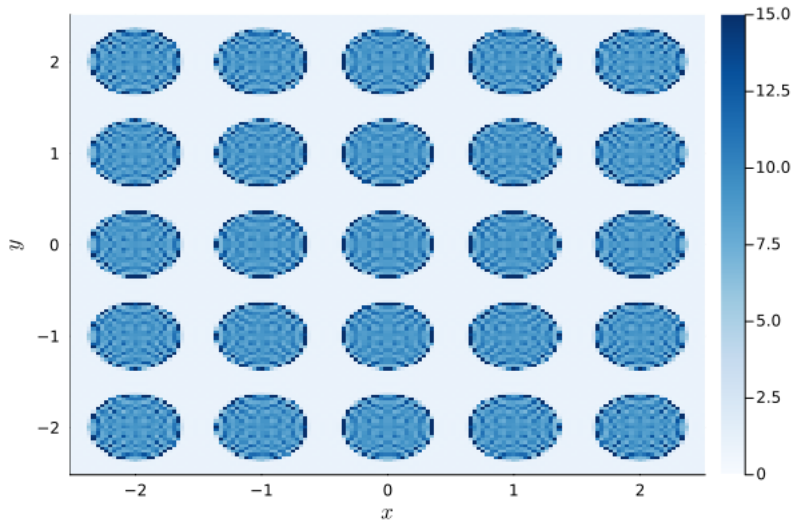

Figure 3. Dielectric functions of the normal section of a photonic crystal square lattice array of infinite dielectric cylinders evaluated numerical as a Fourier series with coefficients $\kappa_{n_x^\varepsilon,n_y^\varepsilon}$ as described in (III.38-III.39). The chosen values are $\varepsilon_0 = 1$, $\varepsilon_0 = 9$, $f = 0.4488$, $a = 1$ with $N = 150 \times 150$ number of points of $\mathbf{r}_{||}$ and $m = 2 \times 20 + 1$ iterations of the Fourier series.

```
G(nx, ny) = [(2π/a)*nx, (2π/a)*ny]
```

```
G (generic function with 1 method)
```

```
function κ(nx, ny)

    ac = a^2
    R = sqrt( (f*ac)/π )

    if nx == 0 && ny == 0
        return (1/εc)*f + (1-f)

    else
        return ( (1/εc - 1) * f * 2 * besselj1( norm(G(nx,ny)) * R ) ) / ( norm(G(nx,ny)) * R )

    end

end
```

```
κ (generic function with 1 method)
```

```
H(kx, ky, mx, my, nx, ny) = norm( [kx,ky] + G(nx, ny) ) * κ(mx-nx, my-ny) * norm( [kx,ky] + G(mx, my) )
```

```
H (generic function with 1 method)
```

Figure 4. Numerical implementation of the dispersion relation: Defining functions.

Now, contrarily to the 1D case where have a second order tensor $\hat{H}_{n,m}$, in the 2D case we have a fourth order tensor $H_{n_x,n_y}^{m_x,m_y}$. Because of this, we can't trivially represent H in matrix form. However, if we find a way to represent $H_{n_x,n_y}^{m_x,m_y}$ in matrix form, we could still easily calculate the eigenfrequencies as the eigenvalues of said matrix. So now, our task is to try express the double summation

$$\sum_{n_x}\sum_{n_y} H_{n_x,n_y}^{m_x,m_y} v^{n_x,n_y} \tag{III.40}$$

as a 2D matrix times a 1D vectors. For this we look at the simplest example of just considering the first 3 terms, this

is $n_x = -1, 0, +1$ and $n_y = -1, 0, +1$. Expressing all the terms of the sum explicitly, we obtain

$$\begin{aligned} &H^{m_x,m_y}_{-1,-1} v^{-1,-1} + H^{m_x,m_y}_{-1,0} v^{-1,0} + H^{m_x,m_y}_{-1,1} v^{-1,0} \\ &+H^{m_x,m_y}_{0,-1} v^{0,-1} + H^{m_x,m_y}_{0,0} v^{0,0} + H^{m_x,m_y}_{0,1} v^{0,1} \\ &+H^{m_x,m_y}_{1,-1} v^{1,-1} + H^{m_x,m_y}_{1,0} v^{1,0} + H^{m_x,m_y}_{1,1} v^{1,1} \\ &= \omega^2 v_{m_x,m_y} \end{aligned} \tag{III.41}$$

If we now chose to express the second order tensor v_{m_x,m_y} as a vector (instead of a matrix) like

$$\mathbf{v} = \left[v^{-1,-1}, v^{-1,0}, v^{-1,1}, v^{0,-1}, v^{0,0}, v^{0,1}, v^{1,-1}, v^{1,0}, v^{1,1}\right] \tag{III.42}$$

we can express (III.26) in matrix form, for all the terms m_x, m_y, as

$$\begin{bmatrix} H^{-1,-1}_{-1,-1} & H^{-1,-1}_{-1,0} & H^{-1,-1}_{-1,1} & H^{-1,-1}_{0,-1} & H^{-1,-1}_{0,0} & H^{-1,-1}_{0,1} & H^{-1,-1}_{1,-1} & H^{-1,-1}_{1,0} & H^{-1,-1}_{1,1} \\ H^{-1,0}_{-1,-1} & H^{-1,0}_{-1,0} & H^{-1,0}_{-1,1} & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ H^{-1,1}_{-1,-1} & H^{-1,1}_{-1,0} & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ H^{0,-1}_{-1,-1} & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ H^{0,0}_{-1,-1} & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ H^{0,1}_{-1,-1} & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ H^{1,-1}_{-1,-1} & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ H^{1,0}_{-1,-1} & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ H^{1,1}_{-1,-1} & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \end{bmatrix} \begin{bmatrix} v^{-1,-1} \\ v^{-1,0} \\ v^{-1,1} \\ v^{0,-1} \\ v^{0,0} \\ v^{0,1} \\ v^{1,-1} \\ v^{1,0} \\ v^{1,1} \end{bmatrix} = \frac{\omega^2}{c^2} \begin{bmatrix} v_{-1,-1} \\ v_{-1,0} \\ v_{-1,1} \\ v_{0,-1} \\ v_{0,0} \\ v_{0,1} \\ v_{1,-1} \\ v_{1,0} \\ v_{1,1} \end{bmatrix} \tag{III.43}$$

If you can't see right away why this is the case, consider, for example, the isolated case for $m_x = -1$ and $m_y = -1$. From (III.26) we have that

$$\sum_{n_x} \sum_{n_y} H^{-1,-1}_{n_x,n_y} v^{n_x,n_y} = \frac{\omega^2}{c^2} v_{-1,-1} \tag{III.44}$$

Making all the terms explicit, we obtain

$$\begin{aligned} &H^{-1,-1}_{-1,-1} v^{-1,-1} + H^{-1,-1}_{-1,0} v^{-1,0} + H^{-1,-1}_{-1,1} v^{-1,0} \\ &+H^{-1,-1}_{0,-1} v^{0,-1} + H^{-1,-1}_{0,0} v^{0,0} + H^{-1,-1}_{0,1} v^{0,1} \\ &+H^{-1,-1}_{1,-1} v^{1,-1} + H^{-1,-1}_{1,0} v^{1,0} + H^{-1,-1}_{1,1} v^{1,1} \\ &= \omega^2 v_{-1,-1} \end{aligned} \tag{III.45}$$

Which is exactly the same as the first term of the matrix-vector multiplication in (III.43) since, doing the dot product of the first row of the matrix by the column vector yields

$$\begin{bmatrix} H^{-1,-1}_{-1,-1} & H^{-1,-1}_{-1,0} & H^{-1,-1}_{-1,1} & H^{-1,-1}_{0,-1} & H^{-1,-1}_{0,0} & H^{-1,-1}_{0,1} & H^{-1,-1}_{1,-1} & H^{-1,-1}_{1,0} & H^{-1,-1}_{1,1} \end{bmatrix} \begin{bmatrix} v^{-1,-1} \\ v^{-1,0} \\ v^{-1,1} \\ v^{0,-1} \\ v^{0,0} \\ v^{0,1} \\ v^{1,-1} \\ v^{1,0} \\ v^{1,1} \end{bmatrix} = \frac{\omega^2}{c^2} v_{-1,-1}$$

If we now repeat this process for the other values of m_x and m_y we construct the matrix exactly as in (III.43).

Thus, the next numerical implementation step if to construct our matrix just like (III.43) but generic for all matrix sizes. This is shown in Fig (5). In this piece of code we start be defining a row vector with dimension $(2S+1)^2$ and then, for a given $\mathbf{k} = (k_x, k_y)$, we assemble the matrix one entry $H^{m_x,m_y}_{n_x,n_y}$ at a time, one row at a time. To do this,

we start by setting $m_x = -S$ and $m_y = -S$ to build the corresponding first row and then, if we repeat the process $(2S+1)^2$ times running for all m_x and m_y and concatenating all rows vertically along the way, we obtain our matrix of dimensions $(2S+1)^2 \times (2S+1)^2$. Note that, the last command in **return** in the line [24] is just to delete the first empty row we used to define the matrix size in the beginning.

```
1  function MH2D(kx, ky, S)
2
3      MH2D = zeros(Float64, 1, (2S+1)^2)
4
5      for mx in range(-S, S, step = 1)
6          for my in range(-S, S, step = 1)
7
8              Rows = []
9
10             for nx in range(-S, S, step = 1)
11                 for ny in range(-S, S, step = 1)
12
13                     Hs = copy( H(kx, ky, nx, ny, mx, my) )
14                     Rows = push!(Rows, Hs)
15
16                 end
17             end
18
19             Rows = transpose(Rows)
20             MH2D = [MH2D; Rows]
21
22         end
23     end
24
25     return real( MH2D[1:end .∉ 1, 1:end] )
26
27 end
```

```
MH2D (generic function with 1 method)
```

Figure 5. Numerical implementation of the dispersion relation: Assembling the $H^{m_x,m_y}_{n_x,n_y}$ matrix.

Finally, the eigenfrequencies for a given $\mathbf{k} = (k_x, k_y)$ can be readily obtain as the eigenvalues of the built-in matrix this is

$$\omega(\mathbf{k}) = \sqrt{\text{Eigenvalues}\left(\hat{H}(\mathbf{k})\right)} \tag{III.46}$$

as is shown in the first cell of Fig (7). For example, for $\mathbf{k} = \mathbf{0}$ this function would return a vector like

$$\left[\ \omega_{\text{band }1}(0,0)\ \ \omega_{\text{band }2}(0,0)\ \ \omega_{\text{band }3}(0,0)\ \ \ldots\ \right] \tag{III.47}$$

where the successive eigenvalues correspond to the value of $\omega(\mathbf{k})$ in the successive bands. Basically, what we are doing is building the band structure one $\mathbf{k}$ at a time.

Therefore, the next and final step, is to calculate the eigenfrequencies $\omega(\mathbf{k})$ for all $\mathbf{k}$. This is done by going along the edge of the irreducible Brillouin zone following the usual path $\mathbf{k} : \Gamma \to X \to M \to \Gamma$, as shown in Fig (6). The reasoning for choosing such a path to visualize the bandgaps is that all the frequencies corresponding to the interior points of a first Brillouin zone typically fall in between the lowest and the highest frequencies of the states at the irreducible zone edge. As the goal, frequently, is to establish the presence of a complete bandgap (a bandgap for all transverse propagation directions) then one can do that just by checking the points of the Brillouin zone edge.

Finally, in the second cell of Fig (7) we assemble all the eigenfrequencies $\omega(\mathbf{k} : \Gamma \to X \to M \to \Gamma)$ in a matrix such that, in each row we have the eigenfrequencies of all the band for a fixed $\mathbf{k}$ and in each column we have the

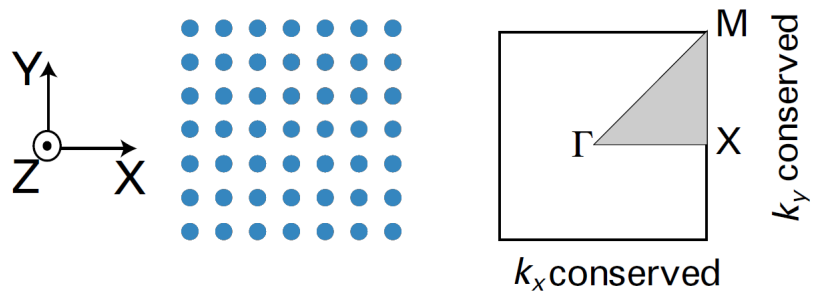

Figure 6. Path along the edge of the irreducible Brillouin zone $\mathbf{k} : \Gamma \to X \to M \to \Gamma$ given the crystal symmetries.

eigenfrequencies of all the $\mathbf{k}$'s for a fixed band. To illustrate the idea, we will obtain something like

$$\begin{bmatrix}
\omega_{\text{band 1}}(0.0,0.0) & \omega_{\text{band 2}}(0.0,0.0) & \omega_{\text{band 3}}(0.0,0.0) & \dots \\
\omega_{\text{band 1}}(0.1,0.0) & \omega_{\text{band 2}}(0.1,0.0) & \omega_{\text{band 3}}(0.1,0.0) & \dots \\
\vdots & \vdots & \vdots & \dots \\
\vdots & \Gamma \to X : k_x = 0.0 \to k_x = \pi/a & \vdots & \dots \\
\vdots & \vdots & \vdots & \dots \\
\omega_{\text{band 1}}(\pi/a,0.0) & \omega_{\text{band 2}}(\pi/a,0.0) & \omega_{\text{band 3}}(\pi/a,0.0) & \dots \\
\omega_{\text{band 1}}(\pi/a,0.1) & \omega_{\text{band 2}}(\pi/a,0.1) & \omega_{\text{band 3}}(\pi/a,0.1) & \dots \\
\vdots & \vdots & \vdots & \dots \\
\vdots & X \to M : k_y = 0.0 \to k_y = \pi/a & \vdots & \dots \\
\vdots & \vdots & \vdots & \dots \\
\omega_{\text{band 1}}(\pi/a,\pi/a) & \omega_{\text{band 2}}(\pi/a,\pi/a) & \omega_{\text{band 3}}(\pi/a,\pi/a) & \dots \\
\omega_{\text{band 1}}(\pi/a-0.1,\pi/a-0.1) & \omega_{\text{band 2}}(\pi/a-0.1,\pi/a-0.1) & \omega_{\text{band 3}}(\pi/a-0.1,\pi/a-0.1) & \dots \\
\vdots & \vdots & \vdots & \dots \\
\vdots & M \to \Gamma : \mathbf{k} = (\pi/a,\pi/a) \to \mathbf{k} = (0,0) & \vdots & \dots \\
\vdots & \vdots & \vdots & \dots \\
\omega_{\text{band 1}}(0.0,0.0) & \omega_{\text{band 2}}(0.0,0.0) & \omega_{\text{band 3}}(0.0,0.0) & \dots
\end{bmatrix} \tag{III.48}$$

Again, to assemble such a matrix, we puzzle together one entry $\omega(\mathbf{k})$ at a time, one row at a time. We start by setting $k_x = 0$ and $k_y = 0$ to build the corresponding first row and then we repeat the process N_k times (where N_k is the number of different k's) running for all k_x and k_y in $\Gamma \to X$, concatenating all rows vertically along the way. We then repeat the process for all k_x and k_y in $X \to M$ and for all k_x and k_y in $M \to \Gamma$, obtaining a matrix of dimensions $3N_k \times (2S+1)^2$. Once gain, the last command in **return** in the line [35] is just to delete the first empty row we used to define the matrix size in the beginning.

```julia
ω(kx, ky, S) = sqrt.( eigvals( MH2D(kx, ky, S) ) )
```

```
ω (generic function with 1 method)
```

```julia
function ωbandas(S, Nks)

    ωs = zeros(ComplexF64, 1, (2S+1)^2)

    for kx in range(0, π/a, length = Nks)

        ΓtoX = ω(kx, 0, S)
        ΓtoX = transpose(ΓtoX)
        ωs = [ωs; ΓtoX]

    end

    for ky in range(0, π/a, length = Nks)

        XtoM = ω(π/a, ky, S)
        XtoM = transpose(XtoM)
        ωs = [ωs; XtoM]

    end

    for kx in reverse( range(0, π/a, length = Nks) )
        for ky in reverse( range(0, π/a, length = Nks) )

            if kx == ky

                MtoΓ = ω(kx, ky, S)
                MtoΓ = transpose(MtoΓ)
                ωs = [ωs; MtoΓ]

            end

        end
    end

    return real( ωs[1:end .∉ 1, 1:end] ) / (2*π)

end
```

```
ωbandas (generic function with 1 method)
```

Figure 7. Numerical implementation of the dispersion relation: Calculating the eigenfrequencies $\omega(\mathbf{k} : \Gamma \to X \to M \to \Gamma)$.

6. *Results*

Printing the results of the various eigenfrequencies $\omega(\mathbf{k} : \Gamma \to X \to M \to \Gamma)$, results in the band structures shown in Fig (8). In this figure we show three different band structures (up to the 10th band) depending on the value of the radii of the dielectric cylinders. In Fig (8)**(a)** we consider the non-limit case where $R = \sqrt{f/\pi}$ with $f = 0.4488$. Here, we can clearly see the appearance of three distinct photonic bandgaps, denoted by the gray regions. Alongside with this case, we also consider the limit cases where **(b)** the diameter of the dielectric cylinders $2R$ approaches the distance a between each other and **(c)** where the radius of the dielectric cylinders R approaches zero. In this two limiting cases, one can see that, while fairly similar to the non-limiting case, the photonic bandgaps fully close.

Notice that, for the special case **(c)**, it seems like the different bands almost snap into place at the points Γ, X and M. Since, for all intents and purposes, the special case **(c)** roughly corresponds to a 2D square grid of atoms, we have a greater deal of symmetries and thus, it is intuitive to see why the band structure converges in the points of higher symmetry Γ, X and M.

Another distinction among the three different regimes is that, as R gets smaller, the eigenfrequencies get bigger. For example, the top of the 10th band for $R = 0.001$ reaches almost $\omega \approx 2$ while, for $R = 0.38$ we have $\omega \approx 0.8$ and, for $R = 0.499$, we have $\omega \approx 0.7$.

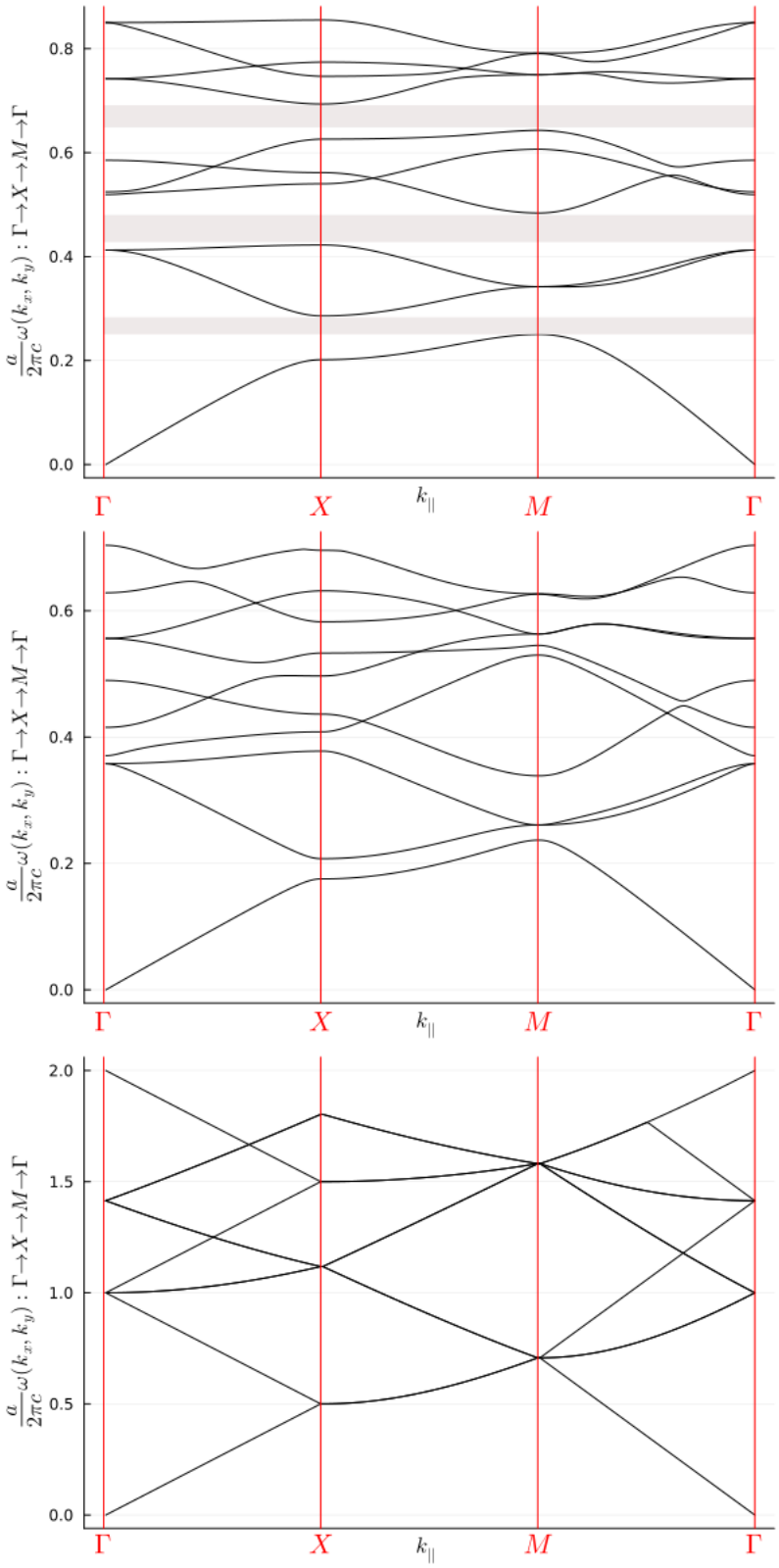

Figure 8. Dispersion relation of the square lattice photonic crystal array. The plot is of $a/2\pi c \times \omega(\mathbf{k}_{||})$ versus $\mathbf{k}_{||}$ where a is the nearest neighbor separation of the lattice sites and c is the speed of light. In the plot, the radius of the dielectric cylinders was taken as **(a)** $R = \sqrt{f/\pi} \approx 0.38$ with $f = 0.4488$ **(b)** $R = 0.499$ and **(c)** $R = 0.01$ and the dielectric constant as $\varepsilon_c = 9$. Also, for the plots, it was used $3N_k = 300$ points.

IV. APPENDIX

A. Jacob-Anger Expansion

The Jacob-Anger expansion represents an expansion of plane waves into a series of cylindrical waves as follows

$$e^{-iz\cos\theta} = \sum_{n=-\infty}^{+\infty} (-i)^n J_n(z) e^{-in\theta} \tag{IV.1}$$

where J_n are Bessel functions of the first kind defined as

$$J_n(z) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!\Gamma(m+n+1)} \left(\frac{z}{2}\right)^{2m+n} \tag{IV.2}$$

and Γ is the gamma function defined as

$$\Gamma(n) = (n-1)! \qquad , \text{for } n \in \mathbb{Z}^+ \tag{IV.3}$$

To proof this expansion one can start by expanding the cosine as its complex exponential form. We obtain

$$e^{-iz\cos\theta} = e^{-iz\left(\frac{e^{i\theta}+e^{-i\theta}}{2}\right)} = e^{-\frac{1}{2}ize^{i\theta}} e^{-\frac{1}{2}ize^{-i\theta}} \tag{IV.4}$$

Substituting the exponential by its the Taylor series expansion

$$e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n \tag{IV.5}$$

we further obtain

$$\begin{aligned} e^{-iz\cos\theta} &= \left(\sum_{n=0}^{\infty} \frac{1}{n!}\left(-\frac{1}{2}ize^{i\theta}\right)^n\right)\left(\sum_{m=0}^{\infty} \frac{1}{m!}\left(-\frac{1}{2}ize^{-i\theta}\right)^m\right) \\ &= \sum_{n=0}^{\infty}\sum_{m=0}^{\infty} (-i)^{n+m} \frac{1}{n!m!}\left(\frac{z}{2}\right)^{n+m} e^{i(n-m)\theta} \end{aligned} \tag{IV.6}$$

Now, if we define a new variable $l = n - m$ (and thus $n + m = l + 2m$), we get

$$e^{-iz\cos\theta} = \sum_{l=-\infty}^{\infty}\sum_{m=0}^{\infty} (-i)^{l+2m} \frac{1}{(l+m)!m!}\left(\frac{z}{2}\right)^{l+2m} e^{il\theta} \tag{IV.7}$$

noting that the summation in l goes needs to go over $]-\infty, +\infty[$ since it has a minimum at $n = 0$ and $m = +\infty$ and a maximum at $n = +\infty$ and $m = 0$. Using the Gamma function definition in (IV.3) we can have

$$\Gamma(l+m+1) = ((l+m+1)-1)! = (l+m)!$$

Also, we can simplify the i term as

$$(-i)^{l+2m} = (-i)^l(-i)^{2m} = (-i)^l((-i)^2)^m = (-i)^l(-1)^m$$

It follows that

$$e^{-iz\cos\theta} = \sum_{l=-\infty}^{\infty} (-i)^l \left(\sum_{m=0}^{\infty} \frac{(-1)^m}{m!\Gamma(m+l+1)}\left(\frac{z}{2}\right)^{2m+l}\right) e^{il\theta}$$

Now, the term in curved brackets is exactly the definition of the Bessel functions of first time as shown in (IV.2). We obtain

$$e^{-iz\cos\theta} = \sum_{l=-\infty}^{\infty} (-i)^l J_l(z) e^{il\theta}$$

If we now rename the mute index $l \to -n$ we have that $(-i)^l = ((-1)i)^{-n} = (-1)^{-n}i^{-n} = (-1)^{-n}(-i)^n$ and thus

$$e^{-iz\cos\theta} = \sum_{n=-\infty}^{\infty} (-i)^n \left((-1)^{-n} J_{-n}(z)\right) e^{-in\theta}$$

Since

$$J_{-n}(z) = (-1)^n J_n(z)$$

Proof:

$$J_{-n}(z) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,(k+(-n))!} \left(\frac{z}{2}\right)^{2k+(-n)}$$

Since the Gamma function is only defined for positive integers we restrict the summation in k to start on n,

$$J_{-n}(z) = \sum_{k=n}^{+\infty} \frac{(-1)^k}{k!\,(k+(-n))!} \left(\frac{z}{2}\right)^{2k+(-n)}$$

If we now do the change of variable $p = k - n$ (and thus the summation starts again on $p = 0$) we obtain

$$\begin{aligned} J_{-n}(z) &= \sum_{p=0}^{+\infty} \frac{(-1)^{p+n}}{(p+n)!p!} \left(\frac{z}{2}\right)^{2p+n} \\ &= (-1)^n \sum_{p=0}^{+\infty} \frac{(-1)^p}{p!\,(p+n)!} \left(\frac{z}{2}\right)^{2p+n} \\ &= (-1)^n J_n(z) \qquad \square \end{aligned}$$

We obtain the desired result

$$e^{-iz\cos\theta} = \sum_{n=-\infty}^{\infty} (-i)^n J_n(z) e^{-in\theta} \qquad \square$$